

Dividend Omission Effects in Black–Scholes–Merton Pricing of Short-Maturity United States Equity Call Options

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Abstract

The effect of dividends on option values is theoretically established; the empirical question for very short maturities is whether a cash dividend actually becomes relevant before expiration and whether its omission is economically material. We study call options on nine U.S. equities observed on 13 archived observation dates between August 29 and September 18, 2025, with a common September 19 expiration. To avoid circularity from option-implied volatility, the primary specification uses bid–ask midpoints as observed premiums, ticker-specific annualized volatilities estimated independently from 20 pre-sample daily log returns ending August 28, 2025 and held fixed over the

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option sample, and known discrete cash dividends only when the ex-dividend date lies inside the remaining option life. A fully documented, ticker-neutral screen for valid quotes, both call no-arbitrage bounds, and recent trading activity retains 4,307 of 9,959 option-day observations. Of these, 1,340 observations have a dividend ex-date before expiration; AAPL, JPM, MSFT, PFE, and WFC have no relevant ex-date and therefore receive no primary dividend adjustment. The discrete-dividend omission effect implied by the benchmark is closely approximated by $\Delta PV(D)$ (correlation 0.999908; mean absolute approximation error \$0.0061). Conditional on dividend relevance, the mean omission effect is \$0.499 versus a mean bid-ask half-spread of \$0.339. The ratio of mean omission effect to mean half-spread is 3.33 for BAC, 2.23 for MRK, 2.13 for UNH, and only 0.02 for NVDA. Thus dividend omission is economically material relative to quoted trading frictions for BAC, MRK, and UNH but not NVDA. Correcting the dividend input does not, however, uniformly improve market-price fit under the independent-volatility benchmark, illustrating that the mechanical valuation effect and empirical pricing accuracy are distinct objects.

1 Introduction

The Black-Scholes framework remains a central benchmark for option valuation because of its tractability and transparent dependence on model inputs [1]. Merton's extension shows that expected dividends reduce the value of a European call relative to an otherwise identical non-dividend-paying underlying [2]. The sign of the dividend effect is therefore not an empirical discovery. For short-maturity options, the more relevant questions are which announced cash dividends actually become ex-dividend before expiration, how large the resulting omission effect is, and whether that magnitude is meaningful relative to market trading frictions.

These distinctions matter because annualized dividend yield can be a poor short-horizon proxy. A stock may have a large annual dividend yield even though its next ex-dividend date lies after the option expires; conversely, a single cash dividend immediately before expiration

can be economically important. The present revision therefore treats known discrete cash dividends as the primary specification and uses continuous dividend yield only as a robustness comparison.

The empirical exercise also separates dividend treatment from other sources of model-market divergence. U.S. single-stock options are American-style while the closed-form Black-Scholes benchmark is European; volatility is not constant; and option quotes can be stale or wide. Further, reusing option-implied volatility as an input and interpreting the resulting fit as independent validation is circular. We consequently use historical volatility estimated from pre-sample underlying returns, construct market premiums from bid-ask midpoints, impose fully documented and ticker-neutral quote/no-arbitrage/activity screens, and assess economic significance relative to bid-ask half-spreads.

Our contribution is therefore narrow and reproducible. First, we quantify the exact price effect of omitting a known cash dividend and compare it with a first-order delta approximation. Second, we establish whether the omission effect is large relative to observed transaction-price uncertainty. Third, we test whether applying the mechanically correct dividend adjustment actually improves agreement with observed market prices under an otherwise simplified pricing benchmark. These are related but distinct questions.

2 Related Work

Black and Scholes [1] derived the closed-form European option-pricing relation under constant volatility, continuous trading, and no dividends, while Merton [2] incorporated a continuous proportional dividend yield. Research on dividend-paying American calls, including Roll [3], Geske [4], and Whaley [5], established the importance of discrete dividends and early-exercise incentives.

Empirical studies have long documented systematic departures from constant-volatility Black-Scholes pricing. MacBeth and Merville [6] reported moneyness-related pricing errors,

Rubinstein [7] documented cross-strike variation in implied volatility, and Dumas, Fleming, and Whaley [8] showed that richer volatility specifications can improve in-sample fit without guaranteeing superior out-of-sample performance. Stochastic-volatility models such as Hull and White [9] and Heston [10], together with the comparisons of Bakshi, Cao, and Chen [11], further demonstrate that volatility misspecification can be quantitatively important.

Short-dated options are useful for isolating local pricing effects but are especially sensitive to microstructure and quote quality [12, 13]. Muravyev and Pearson [14] emphasize that option trading-cost measurement depends materially on the price proxy used. This motivates our use of valid bid–ask midpoints and our explicit comparison of dividend-omission effects with bid–ask half-spreads.

3 Methodology

3.1 Non-dividend benchmark and discrete-dividend adjustment

For a European call on a non-dividend-paying stock, Black and Scholes [1] give

$$C_0 = SN(d_1) - Ke^{-rT}N(d_2), \quad (1)$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (2)$$

For a known cash dividend D whose ex-dividend date occurs τ years after the valuation date and before option expiration, we use the standard spot-adjustment approximation

$$PV(D) = De^{-r\tau}, \quad S_D = S - PV(D), \quad (3)$$

and compute

$$C_D = BS(S_D, K, T, r, \sigma). \quad (4)$$

If no ex-dividend date lies inside the remaining option life, $PV(D) = 0$ and therefore $C_D = C_0$ exactly. The primary analysis contains at most one relevant cash dividend per contract because of the very short remaining maturity.

The exact mechanical omission effect is

$$\Omega = C_0 - C_D. \quad (5)$$

A first-order Taylor approximation gives

$$\Omega \approx \Delta_D PV(D), \quad \Delta_D = N(d_{1,D}), \quad (6)$$

where $d_{1,D}$ is evaluated at the adjusted spot S_D . Equation (6) is an analytical benchmark for the mechanical price effect; it does not imply that C_D must be closer to the observed market premium than C_0 .

3.2 Dividend timing

The option sample expires on September 19, 2025. Since May 2024, normal U.S. cash dividends become ex-dividend on the record date when the record date falls on a business day under the T+1 settlement convention [15]. We therefore identify the relevant ex-date from issuer dividend announcements and histories. The schedule used in the analysis is shown in Table 1. Issuer sources are Apple [16], Bank of America [17], JPMorgan Chase [18], Merck [19], Microsoft [20], NVIDIA [21], Pfizer [22], UnitedHealth Group [23], and Wells Fargo [24].

Table 1: Cash-dividend schedule used for the September 19, 2025 option expiration

Ticker	Ex-date used	Dividend	Relevance to sample
AAPL	Aug. 11, 2025	\$0.26	Before sample; no adjustment
BAC	Sep. 5, 2025	\$0.28	Relevant for snapshots before Sep. 5
JPM	Oct. 6, 2025	\$1.50	After expiration; no adjustment
MRK	Sep. 15, 2025	\$0.81	Relevant for snapshots before Sep. 15
MSFT	Aug. 21, 2025	\$0.83	Before sample; no adjustment
NVDA	Sep. 11, 2025	\$0.01	Relevant for snapshots before Sep. 11
PFE	Jul. 25, 2025	\$0.43	Before sample; no adjustment
UNH	Sep. 15, 2025	\$2.21	Relevant for snapshots before Sep. 15
WFC	Aug. 8, 2025	\$0.45	Before sample; no adjustment

Notes: For normal cash dividends under T+1, the ex-date equals the business-day record date [15]. A dividend enters the option-price adjustment only when the ex-date is strictly after the option snapshot and before September 19, 2025. For JPM, the October 6 date is reported only to document that the next dividend date falls after the September 19 expiration; the dividend was declared during the sample and is not used as an input for any pre-announcement snapshot.

3.3 Data, market premium, and sample construction

Option-chain snapshots were collected from Yahoo Finance for AAPL, BAC, JPM, MRK, MSFT, NVDA, PFE, UNH, and WFC. The 13 archived observation dates are August 29 and September 2–4, 8–12, and 15–18, 2025. September 1 was Labor Day and the market was closed. September 5 was a normal trading day, but the scheduled option-chain acquisition process failed and did not produce a reliable, complete snapshot. We therefore excluded the September 5 output rather than repair, selectively retain, or reconstruct the historical chain ex post. This preserves the rule that the empirical analysis uses only the option-chain snapshots successfully archived by the original acquisition workflow. The saved acquisition files do not establish a more specific reason for that gap, so we do not reconstruct a historical Yahoo option chain ex post. For BAC, whose ex-dividend date is September 5, the

dividend adjustment therefore applies to the available pre-ex-date snapshots (August 29 and September 2–4) and to no later observations. All contracts expire September 19, 2025. The raw files contain 9,959 option-day observations.

The primary observed premium is the bid–ask midpoint,

$$C^{mkt} = \frac{Bid + Ask}{2}. \quad (7)$$

We apply the same filters to every ticker, in the following order. First, bid and ask must be strictly positive, $Ask \geq Bid$, and stock price, strike, and time to expiration must be positive. Second, the market midpoint must satisfy the call upper bound

$$C^{mkt} \leq S. \quad (8)$$

Third, we apply the corresponding discrete-dividend lower quote-consistency bound for the European benchmark,

$$C^{mkt} \geq \max(0, S - PV(D) - Ke^{-rT}). \quad (9)$$

Fourth, because the downloaded option files do not contain a separate timestamp for displayed bid and ask, we use `lastTradeDate` as an activity screen rather than a quote timestamp: the primary sample requires a trade on the snapshot date or no more than one preceding U.S. trading session. The only market holiday relevant to this screen is Labor Day, September 1, 2025.

Table 2 reports every filtering stage. The procedure retains 4,307 observations and is fully ticker-neutral.

Table 2: Reproducible sample construction by ticker

Ticker	Raw	Valid quotes	Upper bound	Lower bound	Activity	Final share
AAPL	1061	865	865	598	482	45.4%
BAC	708	617	617	524	361	51.0%
JPM	839	717	717	522	398	47.4%
MRK	620	368	368	339	262	42.3%
MSFT	1304	1053	1053	825	681	52.2%
NVDA	3037	2660	1885	1424	1002	33.0%
PFE	477	294	294	236	160	33.5%
UNH	1276	831	831	765	662	51.9%
WFC	637	533	533	409	299	46.9%
Total	9959	7938	7163	5642	4307	43.2%

Notes: “Valid quotes” requires positive two-sided quotes with $Ask \geq Bid$ and valid positive S, K, T . Upper and lower bounds are applied sequentially. “Activity” requires `lastTradeDate` within one U.S. trading session of the snapshot.

3.4 NVIDIA split and stale-record audit

NVIDIA implemented a ten-for-one forward stock split effective June 7, 2024, with split-adjusted trading beginning June 10, 2024 [25]. The raw NVDA files contain 3,037 observations; 1,599 have strikes above \$200, and 1,092 of those high-strike rows have `lastTradeDate` before June 10, 2024. These records explain the pre-split-scale strike range seen in the original plots. Retaining these stale pre-split-scale records would mechanically distort normalized moneyness and contaminate model-error summaries because strikes and stale transaction information would not represent contemporaneous September 2025 contracts on a comparable economic basis.

We do not manually divide strikes or premiums by ten because the historical contract-

adjustment status of every stale record cannot be reconstructed from the downloaded files. Instead, the universal filters in Table 2 are applied without reference to ticker identity. NVDA retains 1,002 observations. None of the retained NVDA rows has a `lastTradeDate` before June 10, 2024; the maximum retained strike is \$250 and the maximum retained K/S is 1.435. Thus the revised procedure removes the stale legacy records while retaining contemporaneous high-strike contracts that pass the same rules as every other ticker.

3.5 Time, interest rate, and independent volatility

Time to maturity is computed using ACT/ACT. T is actual calendar days to September 19 divided by 252. The risk-free rate is fixed at $r = 0.0441$ (4.41%) across the short sample, matching the original computational workflow.

The primary volatility parameter is estimated independently of option premiums. For each equity we use 21 daily closing prices from July 31 through August 28, 2025, producing 20 pre-sample log returns,

$$r_t = \ln(S_t/S_{t-1}), \quad \hat{\sigma}_{20} = \sqrt{252} \, sd(r_1, \dots, r_{20}). \quad (10)$$

The estimate is fixed across the 13-observation-date option sample. Ten- and fifteen-return windows are used as robustness checks. The ACT/ACT maturity convention and the $\sqrt{252}$ volatility annualization serve different purposes: T measures the contractual fraction of a calendar year used in discounting and time to expiration, whereas $\sqrt{252}$ annualizes a standard deviation estimated from daily trading-session returns using the conventional approximate number of U.S. trading sessions in one year. Both quantities are expressed on an annual scale; they are not alternative day-count conventions for the same input.

3.6 Error measures, inference, and economic significance

For each retained contract,

$$AE_{0,i} = |C_{0,i} - C_i^{mkt}|, \quad AE_{D,i} = |C_{D,i} - C_i^{mkt}|. \quad (11)$$

Ticker-level mean absolute error (MAE) is reported in dollars, while normalized MAE is

$$NMAE_j = \frac{\sum_{i \in j} |C_i^{model} - C_i^{mkt}|}{\sum_{i \in j} C_i^{mkt}}. \quad (12)$$

To avoid treating many strikes from one chain as independent replicates, absolute errors are averaged within ticker and date before paired t and Wilcoxon signed-rank tests. For tickers without a dividend ex-date inside the option life, $C_D = C_0$ and the paired difference is exactly zero by construction.

Economic significance is assessed using the quoted half-spread

$$h_i = \frac{Ask_i - Bid_i}{2}. \quad (13)$$

We report both the ratio of the mean omission effect to the mean half-spread and the fraction of dividend-relevant contracts for which $\Omega_i > h_i$.

4 Empirical Results

4.1 Cleaned sample

Table 3 summarizes the final 4,307-observation sample. The final mean last-trade age is below one calendar day for every ticker. Relative spreads remain heterogeneous, which motivates evaluating dividend effects relative to half-spreads rather than relying only on dollar magnitudes.

Table 3: Primary sample and inputs

Ticker	Final N	Mean age	Mean rel. spread	Avg. midpoint	HV20	Dividend-relevant N
AAPL	482	0.08	9.0%	36.31	29.3%	0
BAC	361	0.29	8.6%	5.74	23.9%	134
JPM	398	0.20	10.4%	31.74	17.2%	0
MRK	262	0.21	23.2%	5.01	20.9%	188
MSFT	681	0.17	16.1%	38.31	16.4%	0
NVDA	1002	0.14	9.8%	39.30	23.7%	567
PFE	160	0.26	22.3%	1.64	28.4%	0
UNH	662	0.19	25.7%	36.95	52.0%	451
WFC	299	0.21	10.8%	5.27	25.1%	0

Notes: Mean age is calendar days between snapshot date and `lastTradeDate`. Mean relative spread is $100(Ask - Bid)/C^{mkt}$. HV20 is annualized historical volatility from 20 pre-sample returns.

Figure 1 pools all 13 dates and normalizes strike and call value by the contemporaneous stock price. The revised legend and marker styles distinguish observed midpoints, the no-dividend benchmark, and the known-discrete-dividend benchmark.

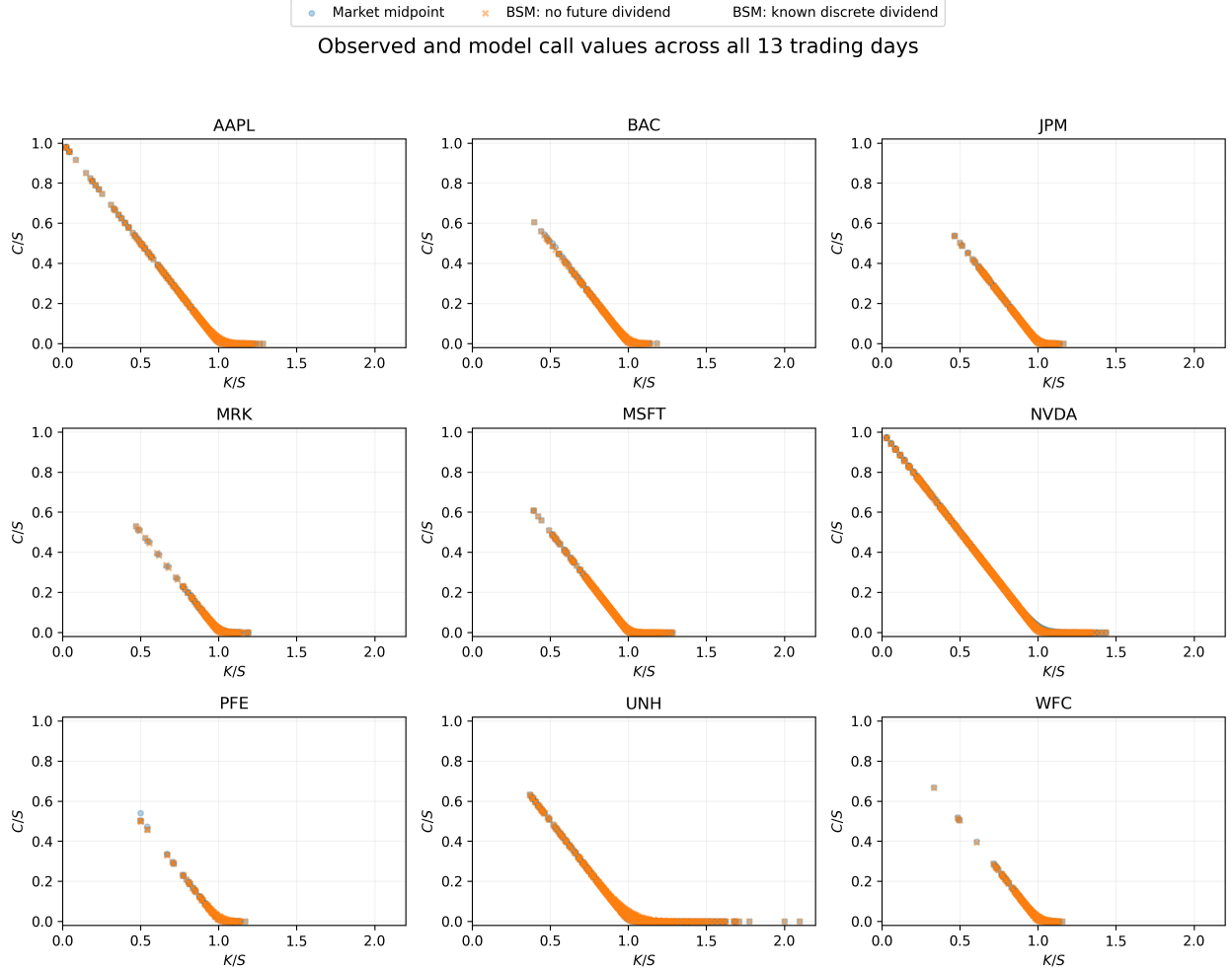


Figure 1: Observed bid–ask midpoints and model values across all 13 trading days. Each panel represents one underlying; strike and call value are normalized by contemporaneous stock price. The discrete-dividend model differs from the no-dividend benchmark only when a known ex-dividend date remains before expiration.

4.2 Mechanical dividend-omission effect

Among the 4,307 retained observations, 1,340 have a known dividend ex-date between the snapshot and expiration. For these dividend-relevant contracts, Equation (6) reproduces the exact omission effect with correlation 0.999908. The mean absolute approximation error is \$0.00608 and the maximum absolute approximation error is \$0.0542. Figure 2 shows the

174 close relationship.

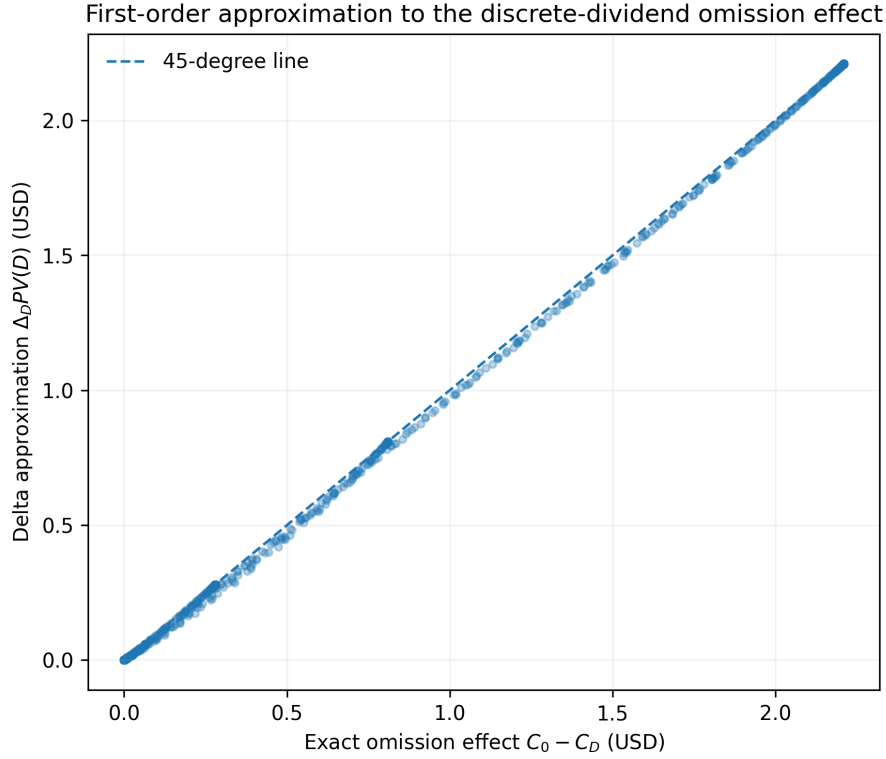


Figure 2: Exact discrete-dividend omission effect $C_0 - C_D$ versus the first-order approximation $\Delta_D PV(D)$ for the 1,340 observations with a dividend ex-date before expiration.

175 Across all 4,307 observations, including the observations for which the correct short-
 176 horizon adjustment is zero, the mean omission effect is \$0.155 and the mean half-spread is
 177 \$0.374. Conditional on dividend relevance, the mean omission effect is \$0.499 and the mean
 178 half-spread is \$0.339, so the ratio of these means is 1.47. This conditional comparison is
 179 the more informative measure of economic significance because observations with no future
 180 ex-date correctly have $\Omega = 0$.

181 Table 4 and Figure 3 show substantial cross-sectional heterogeneity. The mean omission
 182 effect is 3.33 times the mean half-spread for BAC, 2.23 times for MRK, and 2.13 times for
 183 UNH. For NVDA the ratio is only 0.02. The contract-level omission effect exceeds the half-
 184 spread for 93.3% of relevant BAC contracts, 79.8% of relevant MRK contracts, and 73.2%

185 of relevant UNH contracts, but for none of the relevant NVDA contracts.

Table 4: Economic significance of the discrete-dividend omission effect

Ticker	Relevant N	Mean omission	Mean half-spread	Ratio	$\Pr(\Omega > h)$
BAC	134	\$0.209	\$0.063	3.33	93.3%
MRK	188	\$0.422	\$0.189	2.23	79.8%
NVDA	567	\$0.006	\$0.262	0.02	0.0%
UNH	451	\$1.237	\$0.581	2.13	73.2%

Notes: The table is conditional on a known ex-dividend date remaining before September 19, 2025. Ratio is mean omission effect divided by mean half-spread.

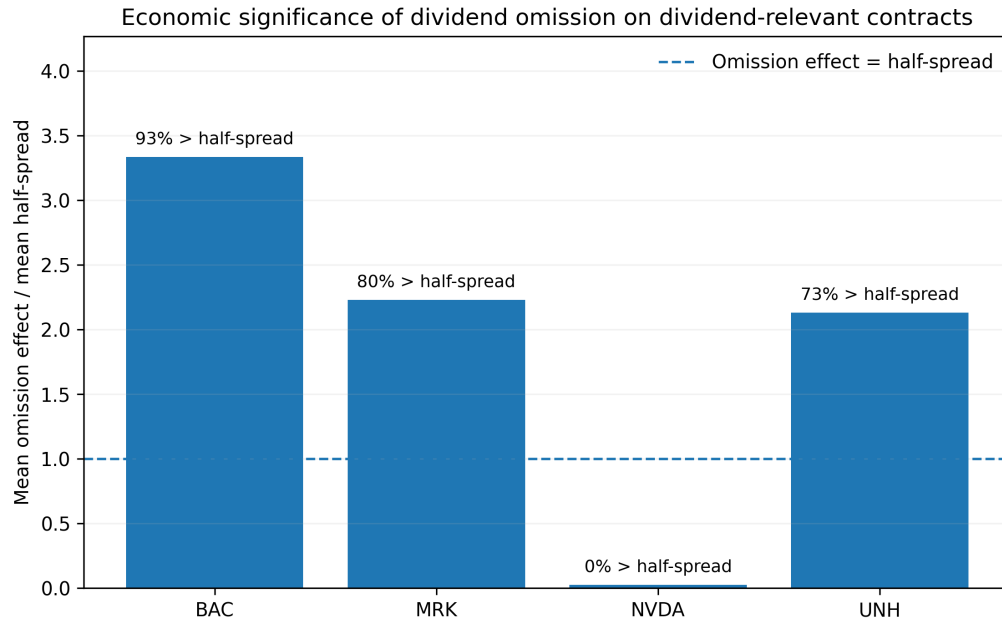


Figure 3: Mean discrete-dividend omission effect scaled by the mean bid-ask half-spread for dividend-relevant contracts. The dashed line at one denotes equality between the two magnitudes; labels report the share of contracts for which the exact omission effect exceeds the individual half-spread.

4.3 Market-price accuracy under independent volatility

Economic materiality of the omission effect does not imply that applying the dividend adjustment must reduce absolute market-pricing error. Table 5 reports the independent-HV20 comparison. For AAPL, JPM, MSFT, PFE, and WFC there is no future ex-dividend date before expiration, so $C_D = C_0$ and all error measures are exactly unchanged. For BAC, MRK, NVDA, and UNH, the discrete-dividend adjustment lowers the theoretical call value. Under the HV20 benchmark, however, pooled MAE does not decline for any of the four dividend-relevant underlyings.

Table 5: Pricing accuracy using independently estimated volatility and known discrete dividends

Ticker	N	MAE C_0	MAE C_D	NMAE C_0	NMAE C_D	MAE change
AAPL	482	0.2531	0.2531	0.697%	0.697%	0.00%
BAC	361	0.0768	0.1000	1.338%	1.743%	-30.32%
JPM	398	0.4798	0.4798	1.512%	1.512%	0.00%
MRK	262	0.1742	0.4132	3.476%	8.246%	-137.24%
MSFT	681	0.4430	0.4430	1.157%	1.157%	0.00%
NVDA	1002	0.3520	0.3555	0.896%	0.905%	-1.01%
PFE	160	0.0550	0.0550	3.354%	3.354%	0.00%
UNH	662	1.2803	1.3083	3.465%	3.541%	-2.18%
WFC	299	0.1092	0.1092	2.074%	2.074%	0.00%

Notes: MAE change is $100(MAE_0 - MAE_D)/MAE_0$; positive values would indicate improved absolute pricing accuracy. Zero changes arise when no ex-dividend date falls inside the remaining option life.

The result is particularly instructive for UNH. Its average signed model error falls from +\$1.057 under C_0 to +\$0.215 under C_D , indicating that the cash-dividend adjustment removes much of the average directional overpricing. Nevertheless, pooled absolute error increases slightly because contract-level errors can cross zero. Thus directional bias, absolute

accuracy, and the mechanical dividend effect should not be conflated.

The paired ticker-day tests in Table 6 are reported only for the four dividend-relevant equities. For other tickers the difference is identically zero. MRK exhibits a statistically detectable deterioration under both tests even after Holm correction. NVDA’s very small deterioration is detectable under the paired t -test after Holm correction but not under the Holm-adjusted Wilcoxon test; its economic omission effect is negligible relative to the half-spread. BAC and UNH do not show statistically detectable differences in daily MAE.

Table 6: Paired daily tests for dividend-relevant underlyings

Ticker	Mean daily $MAE_0 - MAE_D$	t p	Wilcoxon p	Better	Worse	Equal
BAC	-0.0183	0.122	0.125	0	4	9
MRK	-0.2354	< 0.001	0.004	0	9	4
NVDA	-0.0034	0.003	0.016	0	7	6
UNH	-0.0177	0.894	0.910	5	4	4

Notes: Daily errors are first averaged across strikes. “Better/Worse/Equal” counts the 13 sample days according to the dividend-adjusted daily MAE. Holm-adjusted p -values are 0.857/0.875 (BAC), 0.004/0.035 (MRK), 0.024/0.125 (NVDA), and 1.000/1.000 (UNH) for the paired t /Wilcoxon tests, respectively.

4.4 Moneyness-stratified pricing errors

To examine whether dollar MAE is dominated by deep-in-the-money calls, we stratify the cleaned sample by normalized strike K/S : deep ITM ($K/S < 0.80$), ITM ($0.80 \leq K/S < 0.95$), ATM ($0.95 \leq K/S \leq 1.05$), OTM ($1.05 < K/S \leq 1.20$), and deep OTM ($K/S > 1.20$). Table 7 reports both dollar MAE and normalized MAE within each bucket.

Table 7: Pricing errors stratified by moneyness

Moneyiness	N	MAE C_0	MAE C_D	NMAE C_0	NMAE C_D	MAE change
Deep ITM ($K/S < 0.80$)	968	0.2625	0.4193	0.289%	0.462%	-59.73%
ITM ($0.80 \leq K/S < 0.95$)	1131	0.4503	0.4756	1.597%	1.687%	-5.62%
ATM ($0.95 \leq K/S \leq 1.05$)	1156	0.8464	0.7856	18.495%	17.165%	+7.19%
OTM ($1.05 < K/S \leq 1.20$)	799	0.2200	0.1995	103.392%	93.745%	+9.33%
Deep OTM ($K/S > 1.20$)	253	0.0480	0.0458	93.399%	89.114%	+4.59%

Notes: NMAE is normalized by the sum of observed midpoints within the indicated moneyness bucket.

Positive MAE change means lower error under the discrete-dividend specification. The high OTM NMAEs reflect very small option premiums.

The stratification does not support the conjecture that deep-ITM contracts dominate the pooled dollar MAE. Under the no-dividend benchmark, the largest dollar MAE occurs in the ATM bucket (\$0.8464), compared with \$0.2625 for deep ITM calls. Relative error is instead largest for OTM and deep-OTM contracts because their observed premiums are small. The dividend adjustment raises dollar MAE in the deep-ITM and ITM buckets but lowers it in the ATM and OTM buckets, reinforcing the value of reporting moneyness-specific rather than only pooled errors. A ticker-by-moneyness CSV is included in the replication package.

4.5 Robustness and interpretation

Appendix Table 8 reports the numerical robustness results rather than only describing them in text. The primary qualitative conclusions are stable to tighter/looser activity and spread screens: BAC and MRK remain worse under the discrete-dividend adjustment, while NVDA remains economically close to unchanged. UNH is the most specification-sensitive case: the MAE change is +5.11% under HV10 and +1.21% when only same-day activity is retained, but is negative under HV15, HV20, and the other reported screening variants. This sensitivity reinforces our interpretation of historical-volatility pricing as a benchmark rather than an optimal forecast.

The former continuous-yield specification from the submitted revision is retained as a

separate alternative-dividend robustness check in Appendix Table 9. On the new 4,307-observation sample it yields MAE changes of +3.22% for BAC and +4.50% for UNH, while producing negative changes for AAPL, JPM, MRK, MSFT, NVDA, PFE, and WFC. The contrast with the discrete specification demonstrates why annualized yield can be misleading at very short horizons: it imposes a dividend effect even when no ex-date lies before expiration.

The observed contracts are American-style whereas the primary closed-form benchmark is European. We therefore interpret $C_0 - C_D$ only as the mechanical dividend effect within a European benchmark, not as an American-option valuation. Early exercise may matter for deep-in-the-money calls immediately before an ex-dividend date, so residual errors may also reflect exercise style, volatility misspecification, and quote/timing noise. A structural American model is beyond the primary experiment; the replication package provides auxiliary intrinsic-value and escrowed-dividend CRR diagnostics.

5 Discussion and Conclusion

The second-round revision sharpens the empirical question in two ways. First, dividend relevance for a short-dated option depends on calendar timing, not on annualized dividend yield alone. Five of the nine stocks in the sample have no ex-dividend date inside the remaining option life and therefore correctly receive no dividend adjustment. Second, the economic importance of the omission effect should be judged relative to market trading frictions rather than by dollar size in isolation.

The resulting evidence is clear on the mechanical effect. Among the 1,340 contracts with a future ex-dividend date before expiration, the discrete-dividend omission effect is almost perfectly tracked by the first-order $\Delta_D PV(D)$ approximation. More importantly, the effect is economically large relative to quoted half-spreads for BAC, MRK, and UNH: their ratios of mean omission effect to mean half-spread are 3.33, 2.23, and 2.13. NVDA's \$0.01 dividend produces a negligible ratio of 0.02.

Market-price accuracy remains a separate question. Under independently estimated

HV20, applying the known cash dividend does not reduce pooled MAE for the four dividend-relevant names. This does not contradict dividend theory. The cash-dividend correction lowers a call price mechanically; it improves absolute fit only when that movement offsets, rather than reinforces or crosses, other model and data discrepancies. UNH illustrates the distinction: the adjustment sharply reduces average signed overpricing but does not reduce pooled absolute error under HV20.

The revised data audit also resolves the earlier NVDA anomaly. NVIDIA’s June 2024 ten-for-one split is explicitly documented, and the stale pre-split-scale rows are shown to be eliminated by common filters rather than by ad hoc ticker-specific deletions or manual rescaling. The full filter sequence, dividend schedule, raw data, and executable notebook are included in the replication package.

Several limitations remain. The primary volatility input is intentionally simple and fixed over a short sample; the contracts are American-style while the closed-form benchmark is European; and the Yahoo snapshots do not provide a separate timestamp for displayed bid and ask. The discrete-dividend spot adjustment is therefore best interpreted as a transparent short-maturity benchmark rather than a complete structural option-pricing model.

The practical conclusion is correspondingly narrow. For short-maturity equity calls, one should first determine whether a known dividend actually becomes ex-dividend before expiration. When it does, omission can be materially larger than the bid–ask half-spread, as for BAC, MRK, and UNH in this sample. When it does not, imposing an annualized dividend yield creates a short-horizon adjustment that is not tied to an actual dividend event. Distinguishing these cases is essential for both reproducible option-pricing analysis and meaningful statements about economic significance.

A Additional Robustness Tables

Table 8: Robustness of discrete-dividend MAE changes

Specification	N	BAC	MRK	NVDA	UNH
Primary: HV20, activity ≤ 1	4307	-30.32%	-137.24%	-1.01%	-2.18%
HV10, activity ≤ 1	4307	-69.62%	-149.03%	-1.00%	+5.11%
HV15, activity ≤ 1	4307	-65.31%	-146.44%	-0.90%	-3.27%
Same-day activity only	3831	-30.55%	-127.67%	-0.97%	+1.21%
Activity ≤ 2 sessions	4532	-26.83%	-148.84%	-1.03%	-4.73%
Relative spread $\leq 25\%$	3661	-30.98%	-147.71%	-1.01%	-2.59%
Relative spread $\leq 50\%$	3919	-30.46%	-144.63%	-1.01%	-2.29%
Relative spread $\leq 100\%$	4204	-30.32%	-141.59%	-1.01%	-2.21%

Notes: Entries are $100(MAE_0 - MAE_D)/MAE_0$ for the four tickers with a relevant ex-dividend date.

Positive values indicate improved absolute pricing accuracy. The no-dividend tickers are omitted because the discrete adjustment is identically zero.

Table 9: Continuous-yield robustness comparison on the final filtered sample

Ticker	N	MAE $q = 0$	MAE continuous q	MAE change
AAPL	482	0.2531	0.2605	-2.92%
BAC	361	0.0768	0.0743	+3.22%
JPM	398	0.4798	0.5759	-20.04%
MRK	262	0.1742	0.1977	-13.50%
MSFT	681	0.4430	0.4859	-9.67%
NVDA	1002	0.3520	0.3526	-0.17%
PFE	160	0.0550	0.0590	-7.27%
UNH	662	1.2803	1.2227	+4.50%
WFC	299	0.1092	0.1290	-18.05%

Notes: This table reproduces the former annualized continuous-yield treatment on the final sample solely as a robustness comparison. Positive values indicate lower MAE relative to the no-dividend benchmark.

Declarations

Ethical approval

Not applicable. This study did not involve human participants, animals, or clinical data requiring ethical review.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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Author contributions

A.M. developed the core mathematical framework and conducted the empirical data analysis. A.M. and O.I. implemented the computational models and prepared the figures. A.M. and O.I. wrote the manuscript. All authors reviewed and approved the final manuscript.

Data availability

The option-chain inputs used in this study were collected from Yahoo Finance. The project repository is available at <https://github.com/abdallahmaiga10/BSM-option-pricing>. A synchronized revision replication package accompanies this submission and contains the discrete-dividend schedule with source links and executable Python/Colab code designed to reproduce the reported filters, tables, and robustness results using the archived option snapshots and pre-sample closing-price series.

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